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Exercise 10

1. **Silhouette coefficient.**

   Given are 8 two-dimensional points with coordinates shown in Table 1. The clustering result is:

$$G_1 = \{s_1, s_2\}, \qquad G_2 = \{s_3, s_4, s_5\}, \qquad G_3 = \{s_6, s_7, s_8\}.$$

   Calculate the silhouette coefficient for the cluster **G2**. Use Euclidean distance as the distance measure.

Table 1: Point coordinates for the silhouette coefficient task.

ID	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8
x	1	2	5	6	8	5	7	8
y	1	3	5	6	4	1	2	1

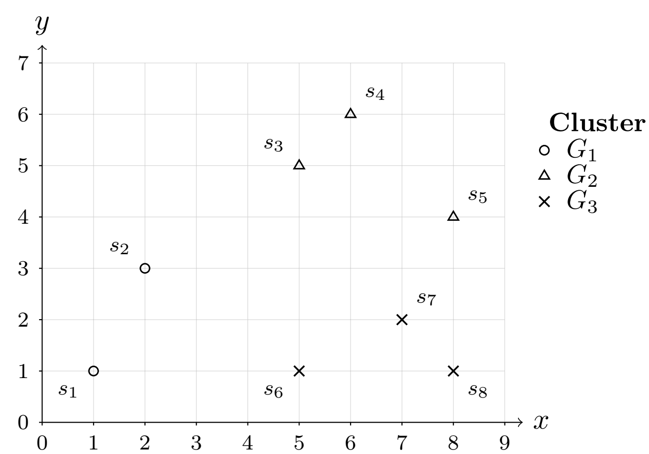

Figure 1: Clustering result for the silhouette coefficient task.

2. Describe the purpose of the feature selection process:
   (a) Why is feature selection done, and what is its goal?
   (b) Which selection methods were presented in the lecture?
   (c) At what point do the presented methods select the final set of features, and to which of the three categories of search methods presented in the lecture do they belong?

3. The quality of selected feature subsets is analyzed in the lecture, among other things, with regard to inter-feature redundancies. Describe the measure and explain:

   (a) To which data types can the measure be applied? How might it need to be modified in order to be applicable?

   (b) In which value range do possible quality values fall? Which value is ideal, and which is catastrophic?

   (c) Describe how you would use the measure and a set of features to assess whether more features should be removed.

4. How can the quality of selected feature subsets be assessed with regard to a performance criterion? Describe and explain the measure that was presented in the lecture:

   (a) To which data types can the measure be applied? How might it need to be modified in order to be applicable?

   (b) In which value range do possible quality values fall? Which value is ideal, and which is catastrophic?

   (c) Describe how you would use the measure and a set of features to assess whether more features should be removed.